Parth Bhandakkar

+91 9131796052 | bhandakkarparth@gmail.com | LinkedIn | GitHub | Portfolio

Experience

AI & Backend Engineer Jan. 2025 – Present
SKDIV | Enterprise AI & Multi-Agent Infrastructure *Remote*

- Architected high-concurrency **FastAPI** LLM orchestration gateways with **Redis** semantic caching, cutting p99 inference latency to **<180ms** at **99.9% uptime**.
- Engineered hybrid dense-sparse **RAG** microservices (**BGE-M3 + BM25** on **Qdrant**), indexing **50,000+ enterprise documents** with automated chunk-level citation verification.
- Deployed stateful multi-agent routing workflows using **LangGraph** and **MCP** tool-calling, reducing manual support ticket escalations by **40%**.
- Built automated evaluation pipelines using **Ragas** and **LangSmith** across a 5-engineer pod, catching retrieval regressions pre-deployment and improving release velocity by **25%**.

AI & Data Engineering Intern Jan. 2024 – Jul. 2024
Devnullx *Remote*

- Engineered partitioned **Apache Kafka** consumer pipelines ingesting real-time market tick feeds into **Pinecone**, sustaining **1,200+ events/sec peak burst** at **<80ms** ingestion latency.
- Implemented two-stage financial **RAG** retrieval with cross-encoder re-ranking, increasing top-5 contextual retrieval precision from 71% to **89%** for downstream forecasting models.
- Deployed **Prometheus** and **Grafana** telemetry dashboards tracking token throughput, p95/p99 latency, and hallucination drift across production LLM endpoints.

Backend Engineer Intern Aug. 2022 – Oct. 2023
WorkZera | B2B Workflow & AI SaaS (Shark Tank India Campus Finalist) *Raipur, India*

- Architected and scaled a multi-tenant **Python/FastAPI** and **PostgreSQL** B2B SaaS backend from zero to **100+ active MSME teams** with role-based access control and background job queues.
- Shipped automated task-delegation agents and semantic workspace search using **LangChain** and vector indexing in 2023, cutting document lookup latency by **40%**.
- Led technical product demos and system architecture reviews, advancing to the national finalist round at **Shark Tank India (University Edition)**.

Projects

Autonomous 24/7 Multi-Agent Job & Outreach Orchestrator | *Python, FastAPI, Gemini, SQLite, Docker* 2026

- Built an autonomous multi-agent daemon running 760+ execution cycles to scrape, score, enrich DNS MX records, compile headless **Tectonic LaTeX** PDFs in 0.15s, and triage IMAP replies.
- Engineered deterministic state-machine guards, rate-limiting jitter, and automated Telegram observability hooks.

Multi-Agent Algorithmic Trading Strategy Builder | *Python, FastAPI, Kafka, InfluxDB, Qdrant* 2026

- Designed an agentic quantitative pipeline combining time-series ingestion (**InfluxDB** + **Kafka**) with semantic news retrieval (**Qdrant**) to backtest and execute automated market strategies.
- Architected scalable **FastAPI** inference microservices handling **50GB+** of daily market telemetry, sustaining sub-**200ms** vector query execution.

Cloud-Native Face Recognition & Photo Retrieval Engine | *FastAPI, FAISS, PyTorch, OpenCV* 2026

- Engineered a sub-100ms vector similarity search backend using **FAISS** and deep metric learning embeddings to index and retrieve high-resolution event photography at scale.
- Containerized GPU-accelerated inference pipelines using **Docker** and asynchronous background workers, cutting batch embedding latency by **45%**.

Technical Skills

Languages & Backend: Python, Go, TypeScript, SQL, FastAPI, Node.js, REST/gRPC, Asyncio, Microservices
AI, LLMs & Agents: LangGraph, LangChain, MCP, RAG, Hybrid Search (BM25 + BGE-M3), vLLM, Ollama, LoRA/PEFT, PyTorch
Data, Streaming & Vector DBs: Qdrant, Pinecone, FAISS, PostgreSQL, Apache Kafka, Redis, InfluxDB
Infra, MLOps & Evaluation: Docker, Kubernetes, AWS, Linux, Git, LangSmith, Ragas, Prometheus, Grafana

Education

International Institute of Information Technology (IIIT), Naya Raipur Naya Raipur, CG
B.Tech in Electronics and Communication Engineering (Co-Author, IEEE Ophthalmic AI Paper) *Dec. 2021 – Jul. 2025*